



SUJET STATAPP 297 - 2026-2027

Sujet proposé par : Kouvabina Irina - irina.kouvabina@engie.com

Organisme : ENGIE

Catégorie : Prédiction

Sujet :

Generate plausible gas consumption scenarios for the upcoming year under unknown weather conditions.

Description du sujet :

Objective: Generate plausible gas consumption scenarios for the upcoming year under unknown weather conditions.

Approach:

? Use a temperature simulator to generate weather scenarios. The true future temperature is unknown, but it is assumed to fall within a given wide range of possibilities.

? Use a gas consumption model that captures the temperature?consumption dependency.

Key Issue:

? The consumption model has residuals. These residuals are resampled and re-injected into the generated scenarios to add realistic variability.

? However, the residuals were estimated from a model calibrated on data with near-normal temperatures, whereas the temperature simulator produces scenarios that include very extreme temperatures.

Consequence / Mismatch:

? Residuals are not representative of behavior under extreme conditions. Injecting them into extreme-temperature scenarios risks producing consumption scenarios that are not plausible ? because the error structure learned from mild conditions may not hold (and likely under- or mis-represents uncertainty) at temperature extremes.

Core Challenge:

How to generate plausible consumption scenarios across the full (including extreme) temperature range when the only available residual information comes from near-normal conditions?

Données du sujet :

gas, temperatures, simulations

Modèles envisagés du sujet :

GAM, GEV

Bibliographie du sujet :

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Notation du sujet :

Commentaires :